

Influence of Sterilization Methods on Selected Soil Microbiological, Physical, and Chemical Properties

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ABSTRACT

Valid results from studies of adsorption of labile organic compounds in soil can be obtained only if microbial degradation of the compound is inhibited. This study investigated effectiveness of sterilization of three silt loams with dry heat, cobalt-60 irradiation, propylene oxide, mercuric chloride, autoclaving, sodium azide, microwave, chloroform, and antibiotics. These treatments were begun after air-dried, sieved soil was moistened to -30 kPa water potential and incubated for 48 h at 25°C . Sterilization was effective and microorganisms were eliminated by cobalt-60 irradiation, propylene oxide, mercuric chloride, and autoclaving $2\times$ or $3\times$ as evidenced by a lack of microbial growth on potato-glucose agar, plate count agar, and nutrient broth. Soil physical and chemical properties were altered by most sterilization treatments. Measured surface areas of the three soils were significantly reduced by propylene oxide. Propylene oxide and sodium azide produced an average pH increase for the three soils of 0.7 and 0.3 units, respectively. All effective sterilization methods except mercuric chloride significantly increased extractable Mn levels in the three soils. None of the effective treatments significantly influenced cation exchange capacity or levels of extractable Ca, Mg, and K of the soils. Mercuric chloride resulted in effective sterilization with minimal changes in soil chemical and physical properties. Following proper safeguards, mercuric chloride could be used to prevent microbial degradation in studies to evaluate adsorption of labile organic chemicals by soil.

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MICROBIAL DEGRADATION of labile organic compounds in soil adsorption studies may lead to erroneous results if a radiolabeled compound and the batch equilibration technique are used (Koskinen and Cheng, 1982; Scott et al., 1982; Scott et al., 1983). Since the amount of radioactivity in the equilibration solution is measured, any labeled degradation product such as CO_2 or metabolites in solution would lead to an underestimation of adsorption. Adsorption or precipitation of metabolites or degradation products would result in an overestimation of adsorption. It is also desirable to distinguish biological from chemical degradation. Historically, this distinction has been made by comparing degradation rates in nonsterilized and sterilized soil (Fletcher and Kaufman, 1980).

Complete soil sterilization destroys all living microorganisms. The most commonly used method of determining the number of microorganisms surviving a given sterilization treatment is plate counts of viable-cell numbers (Korczynski, 1981).

Numerous sterilization procedures have been utilized to eliminate microbial activity in soil, but these procedures often alter soil chemical and physical properties. Skipper and Westermann (1973) reported that soil sterilization with propylene oxide resulted in smaller changes in soil chemical properties than with either sodium azide or autoclaving. Chemical degradation of amitrole in soil was greatly reduced by autoclaving soil because it destroyed the free radical-generating mechanisms required for amitrole degradation (Kaufman et al., 1968). Fletcher and Kaufman

(1980) studied the degradation of 3-chloroaniline in nonsterile and autoclaved soil and that treated with potassium azide, mercuric chloride, or ethylene oxide. They reported that the sterilization procedures all affected degradation, volatilization, and adsorption differently.

Adsorption of organic chemicals by soil has also been shown to be influenced by sterilization techniques. Dao et al. (1982) reported that oven drying, propylene oxide treatment, autoclaving, and cobalt-60 irradiation altered soil adsorption of aniline and diuron compared to air-dried soil.

The purpose of this study was to investigate the influence of various sterilization methods on selected soil microbiological, physical, and chemical properties.

MATERIALS AND METHODS

The procedure for collecting and processing the soil samples in this study was that typically followed for adsorption studies. Captina, Hagerstown, and Palouse silt loams were collected in bulk samples from the field, air-dried, crushed to pass a 495- μ m sieve, and stored in glass containers.

The Palouse silt loam had the largest specific surface area, cation exchange capacity, percent clay, and organic C content of the three soils studied (Table 1). All soils had >50% silt and had salt pH values from 5.4 to 7.0.

To evaluate soil sterilization methods, 2.0 g of each soil was added to sterile 20 by 125 mm screw-top culture tubes, adjusted to -30 kPa water potential with sterile water, and incubated 48 h at 25 °C. Then each treatment was administered. The soil treatments used to evaluate sterilization efficiency included: (i) *air dry*: soil that served as a control; (ii) *oven-dry*: soil samples were dried for 48 h at 90 °C in a forced-draft oven; (iii) *cobalt-60 irradiation*: samples were irradiated with 0.05 to 0.06 MGy with a 0.44 PBq ⁶⁰Co source at the University of Nebraska, Lincoln (McLaren, 1969); (iv) *propylene oxide*: propylene oxide was added at the rate of 7.15 mol/kg soil, and the culture tube was sealed for 48 h and then vented for 48 h (Bartlett and Zelazny, 1967); (v) *mercuric chloride*: the soil was amended with mercuric chloride at a rate of 1.84 mmol/kg (Fletcher and Kaufman, 1980); (vi) *autoclave 3×*: samples were autoclaved at 103.4 kPa and 121 °C for 2 h on Days 1, 2, and 4; (vii) *autoclave 2×*: samples were autoclaved at 103.4 kPa and 121 °C for 1 h, incubated 24 h, and autoclaved an additional 1 h (Ferriss, 1984); (viii) *autoclave 1×*: samples were autoclaved for 2 h at 103.4 kPa and 121 °C (Skipper and Westermann, 1973); (ix) *sodium azide*: 3.08 mmol of sodium azide/kg soil was added and incubated for 72 h at 25 °C (Kaufman et al., 1968); (x) *microwave*: samples were treated for 10 min at 2.45 GHz in a model 8240 Montgomery Wards microwave oven (Montgomery Wards, Chicago), sterile water was added to bring the samples to -30 kPa water potential, and samples were incubated 24 h and then treated for an additional 10 min in the microwave oven (Ferriss, 1984); (xi) *chloro-*

form: samples were treated with 12.3 mol of chloroform/kg soil, placed under vacuum, incubated 24 h, and subsequently treated with an additional 12.3 mol of chloroform/kg soil and incubated under vacuum for 24 h (Jenkinson and Powlson, 1976); (xii) *antibiotics*: samples were amended with 1.0 g of streptomycin sulfate/kg soil plus 7.1 mmol of actidione/kg soil (Anderson and Domsch, 1973).

Soil sterilization efficiency was evaluated with serial dilutions and pour plate techniques. The microbial populations of the soils were determined on moist samples taken from the tubes immediately following completion of the various treatments. The media used were plate count agar for bacterial numbers and potato-glucose agar amended with 0.033 g/L rose bengal and 0.03 g/L streptomycin sulfate for fungal numbers (Wollum, 1982). All plates for bacterial numbers were incubated 7 d and plates for fungal numbers were incubated 4 d at 25 °C prior to counting colonies. Each plate count was a mean of three replications. Additionally, microbial growth was evaluated by adding 10 mL of nutrient broth to the 2.0 g of each treated soil sample immediately following completion of the treatment (Wollum, 1982). The tubes were incubated 7 d at 25 °C, and growth was visually evaluated by formation of turbid medium. The influence of treatments on microbial numbers was evaluated in three replications.

To assess the influence of the various soil sterilization procedures on soil chemical properties, treated samples were analyzed for changes in cation exchange capacity by ammonium saturation (Chapman, 1965). Soil pH was determined in a mixture of 1:1 soil/0.01 M CaCl₂ using a glass electrode (McLean, 1982). Extractable Al, Cd, Cu, Fe, Mn, Ni, Na, and Zn levels were determined by the procedure of Baker (1977). The small exchange soil test procedure of Baker (1980) was also used to evaluate Ca, Mg, and K levels and pH in the soils studied. To evaluate changes in soil physical properties, specific surface area was determined by the ethylene glycol monoethyl ether adsorption procedure (Mortland and Kemper, 1965). The particle size analysis was determined by the hydrometer method (Day, 1965) and the organic C levels were determined using a dry combustion procedure (Nelson and Sommers, 1982).

A minimum of two replications of all physical and chemical measurements was conducted, and statistical analyses of the data included the general linear model procedure (SAS Inst., 1979) and mean separation using the LSD test at the 0.05 level (Snedecor and Cochran, 1980). Because there were significant soil $\times$ treatment interactions for all parameters measured, the data are separated by soil. The statistical analysis of microbiological data was conducted on the log₁₀ transformed data of the form $n + 1$ and presented as geometric means (Baker, 1985).

RESULTS AND DISCUSSION

Sterilization Effectiveness

To prevent microbial degradation of labile organic compounds in an adsorption study, elimination of the

Table 1. Selected physical and chemical properties of the three soils studied.

Soil and location	Classification	Specific surface area m ² /kg $\times 10^3$	Particle size analysis			pH in 0.01 M CaCl ₂	CEC cmol _e /kg	Organic C content %
			Sand	Silt	Clay			
Captina sil Washington Co., AR	Typic Fragiudult	28	33	55	12	5.4	5.9	0.84
Hagerstown sil Frederick Co., MD	Typic Hapludalf	39	28	52	20	7.0	10.0	1.42
Palouse sil Whitman Co., WA	Haploxeroll	116	17	59	24	5.4	22.5	2.17

organisms capable of degrading the compound would be required. Of the various soil treatments utilized, cobalt-60 irradiation, propylene oxide, mercuric chloride, autoclaving 3 $\times$, and autoclaving 2 $\times$ eliminated the microbial population in each of the three soils as determined by plate counts and growth in nutrient broth (Table 2). In the three soils evaluated, autoclaving once for 2 h reduced bacterial numbers by approximately 10⁵ and fungal numbers by approximately 10³ but did not eliminate the microbial population. Compared with air-drying, oven-drying at 90 °C for 24 h significantly reduced the microbial populations in each soil. However, the soils were not sterilized. Labeda et al. (1975) also reported that soil was not sterilized by 3 h of heat treatment at 160 °C.

Sodium azide treatment significantly reduced the bacterial population in the three soils and the fungal population in the Captina and Hagerstown soils.

Microwave treatment under the conditions of this study did not significantly reduce the microbial populations. Vela et al. (1976) showed that soil microorganisms were more susceptible to microwave irradiation in moist soil than in dry soil and that fungi were more susceptible than bacteria. In subsequent work, Vela and Wu (1979) reported various bacteria, actinomycetes, and fungi were inactivated by microwave irradiation when water was present. The lethal effect was due to the increased temperature that resulted when the water absorbed the microwave irradiation. Wainwright et al. (1980) reported that a 20-s exposure of three soils to a 2.45 GHz microwave irradiation had little influence on heterotrophic bacterial numbers, but dramatically reduced fungal counts. Similar findings were reported by Speir et al. (1986). However, wet tissue culture vessels that were artificially contaminated with nine bacterial cultures were shown to be sterilized by a 3-min exposure in a microwave oven (Sanborn et al., 1982). Our results suggest that the amount of water present in the soils and, thus, the temperature increase upon irradiation were insufficient to result in thermal death of the soil microorganisms and spores present.

Chloroform significantly reduced fungal numbers but did not affect the bacterial population. Mineralization of C from soil microorganisms following chloroform fumigation has been used by Jenkinson and Powlson (1976) to determine soil biomass levels. Sev-

eral researchers have reported that chloroform fumigation reduced the microbial population but did not eliminate all microorganisms (Jenkinson and Powlson, 1976; Kassim et al., 1981; Shields et al., 1974).

Soil amendment with streptomycin and/or actidione has been used to separate bacterial and fungal contributions to the total soil respiration (Anderson and Domsch, 1973). Plate counts showed that in the three soils studied and at the antibiotic levels tested, neither the fungal nor the bacterial numbers were significantly reduced by the addition of the antibiotics.

The seven most effective and commonly used treatments, which significantly reduced the microbial population of the three soils and the air-dry and oven-dry treatments, were selected for additional evaluation and comparison.

Soil Surface Area

Oven-drying; cobalt-60 irradiation; mercuric chloride; autoclaving 3 $\times$, 2 $\times$, or 1 $\times$; and sodium azide did not result in a significant change in the surface area of the Captina or Palouse soils (Table 3). However, autoclaving 3 $\times$ or 2 $\times$ reduced the surface area of the Hagerstown soil. Jenneman et al. (1986) presented scanning electron micrographs that demonstrated that typical clay particles, which were plate-like with irregularly shaped edges, became smooth with rounded edges and clumped when autoclaved. They suggested that such a change could result in a reduction in surface area. Propylene oxide may have prevented the binding of the ethylene glycol monoethyl ether used in the surface area determination, which could have resulted in artificially low values in the three soils.

Soil Chemical Properties

Soil pH

Propylene oxide significantly increased the pH of the extraction solution in the Captina, Hagerstown, and Palouse soils by 0.6, 0.3, and 1.1 units, respectively (Table 3). The largest increase in pH was observed in the Palouse soil, which had the highest clay and organic matter content and the lowest pH. Bartlett and Zelazny (1967) studied the influence of propylene oxide on 16 soils and reported an average pH increase

Table 2. Influence of the treatments on the microbial population of the three soils studied.

Treatment	Bacteria			Fungi		
	Captina	Hagerstown	Palouse	Captina	Hagerstown	Palouse
	log ₁₀ colony-forming units/g dry soil					
Air-dry	6.43	6.93	6.99	3.52	4.62	4.49
Oven-dry	2.54	2.28	2.95	2.48	2.65	1.72
Cobalt-60	0	0	0	0	0	0
Propylene oxide	0	0	0	0	0	0
Mercuric chloride	0	0	0	0	0	0
Autoclave 3 $\times$	0	0	0	0	0	0
Autoclave 2 $\times$	0	0	0	0	0	0
Autoclave 1 $\times$	1.43	1.31	1.23	0.96	1.00	1.52
Sodium azide	6.07	6.20	6.33	3.21	4.19	4.37
Microwave	6.07	6.87	6.63	3.26	4.40	4.24
Chloroform	6.54	6.68	6.77	3.19	4.22	4.10
Antibiotics	6.30	6.92	6.92	3.65	4.70	4.58
LSD	0.26	0.32	0.29	0.27	0.26	0.28

Table 3. Influence of the treatments on the surface area and pH of the three soils studied.

Treatment	Surface area			pH in extraction solution		
	Captina	Hagerstown	Palouse	Captina	Hagerstown	Palouse
	m ² /kg			units		
Air-dry	27.7	39.0	115.7	6.5	7.2	6.1
Oven-dry	24.7	30.0	107.7	6.6	7.1	5.9
Cobalt-60	30.3	40.0	123.7	6.9	7.3	6.2
Propylene oxide	17.3	25.3	64.4	7.1	7.5	7.2
Mercuric chloride	25.7	39.3	115.7	6.4	7.0	5.6
Autoclave 3×	21.0	23.3	108.4	6.7	7.2	5.8
Autoclave 2×	23.0	24.0	105.4	6.7	7.2	6.0
Autoclave 1×	24.3	33.0	111.4	6.8	7.2	6.1
Sodium azide	27.0	38.3	120.0	7.0	7.4	6.2
LSD	10.3	9.1	15.1	0.1	0.1	0.1

of 0.4 units with the largest pH increases occurring in acid soils high in organic matter. Skipper and Westermann (1973) also reported a 0.5 to 1.1 unit increase in pH following propylene oxide addition to the three soils they studied, and Clark (1950) noted pH increases of 1.05 and 1.84 units in two soils treated with ethylene oxide.

The toxicity of propylene oxide is apparently related to esterification of the carboxyl groups in protein (Fraenkel-Conrat, 1944). The reaction mechanism would also account for the observed pH increase in propylene oxide-treated soils (Fig. 1). The dissociable protons from carboxyl and hydroxyl groups in organic matter and clay could become neutralized during the reaction with propylene oxide. The results of Bartlett and Zelazny (1967) and those from our study are compatible with the mechanism in that the largest pH increase was noted in the soil with the highest organic matter level and the lowest pH.

Sodium azide treatment significantly increased the pH in the Captina and Hagerstown soils. Parochetti and Warren (1970) also noted a significant pH increase in soils treated with 200 mg/kg potassium azide. Azide has been shown to be converted to hydrazoic acid in soil, resulting in a substantial pH increase (Rozyski and Bartha, 1981). The lack of increase in pH in the Palouse soil was most likely related to the higher buffering capacity of the Palouse compared to the Captina and Hagerstown soils. Because of the increase in pH, which resulted from both the propylene oxide and sodium azide treatments, neither material would be a suitable microbial inhibitor in soil studies where a constant pH is required. The remaining treatments resulted in pH changes of ≤ 0.5 units for the three soils.

Soil Organic Matter Functional Groups

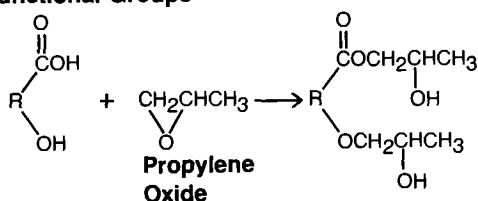


Fig. 1. Reaction of propylene oxide with carboxyl and phenolic hydroxyl groups present in soil organic matter. The reaction presented is a modification of that presented by Fraenkel-Conrat (1944) and discussed by Bartlett and Zelazny (1967).

Extractable Mn, Al, and Fe

The extractable Mn levels of the soils were greatly influenced by the treatments (Table 4). The autoclaving 3×, 2×, or 1× treatments increased the Mn levels in the soils by an average of approximately 500% with the Hagerstown exhibiting the largest increase. Several researchers have reported large increases in Mn levels in autoclaved soils (Fujimoto and Sherman, 1948; Martin et al., 1973; Sonneveld and Voogt, 1973; Skipper and Westermann, 1973). Bohn (1970) studied nine soils and showed that autoclaving for 15 min in 0.01 M CaCl₂ resulted in pH increases in eight of the soils with Mn²⁺ increases of three to >42-fold. He also showed that increasing the autoclaving time from 15 to 240 min did not result in increased Mn²⁺ levels in soil solution for four soils studied. He suggested that the solubility of MnO₂ controlled the equilibrium concentration of Mn²⁺, which he described by the relationship $\text{pH} - \frac{1}{2} \text{pMn}$.

As previously reported by Fujimoto and Sherman (1945), oven-drying the soils also resulted in significant increases in Mn levels. Fujimoto and Sherman (1945) suggested that the increase in Mn was due to dehydration of manganese oxides at temperatures >70 °C. An increase in Mn levels in each soil was also observed with sodium azide treatment and cobalt-60 irradiation, although to a lesser extent than with the heat treatments. The increase in Mn levels occurred even though the soil pH values generally increased. McLaren (1969) also noted the increase in Mn following gamma irradiation of soil, and Skipper and Westermann (1973) reported that sodium azide and propylene oxide increased extractable Mn levels in three soils studied. The only treatment that did not significantly increase Mn in the three soils was mercuric chloride.

In the Captina and Palouse soils, none of the treatments resulted in extractable Al values that were significantly different from those of the air-dried soils (Table 4). In the Hagerstown soil, autoclaving and propylene oxide resulted in significant increases in Al levels compared to levels in the air-dried treatment. The reason for the increase is unknown.

Extractable Fe was significantly reduced only by the sodium azide treatment when compared to the air-dried Hagerstown soil (Table 4). In the Captina, cobalt-60 irradiation and mercuric chloride were the only sterilization methods that did not significantly reduce

Table 4. Influence of the treatments on the extractable Mn, Al, and Fe levels of the three soils studied.

Treatment	Extractable Mn			Extractable Al			Extractable Fe		
	Captina	Hagerstown	Palouse	Captina	Hagerstown	Palouse	Captina	Hagerstown	Palouse
	mg/kg								
Air-dry	68.6	41.4	60.6	11.9	4.6	6.3	25.4	12.2	66.2
Oven-dry	288.5	221.5	182.6	1.5	6.7	1.1	5.9	7.4	50.8
Cobalt-60	133.0	81.7	133.1	19.9	9.1	3.0	24.8	10.1	42.4
Propylene oxide	83.0	58.1	66.1	9.7	16.0	1.4	14.6	13.0	29.0
Mercuric chloride	66.7	49.6	58.7	9.6	4.5	4.3	27.5	11.5	77.9
Autoclave 3×	329.5	311.0	210.6	1.7	11.4	1.0	10.0	8.5	44.1
Autoclave 2×	334.5	305.0	215.1	10.8	15.0	0.7	11.9	13.3	35.3
Autoclave 1×	316.0	251.5	210.1	20.0	16.6	2.3	13.6	10.8	25.4
Sodium azide	241.0	162.0	155.6	15.1	4.6	1.5	12.7	6.4	37.8
LSD	11.3	15.3	7.8	17.3	5.1	NS	6.2	5.1	4.8

extractable Fe. Mercuric chloride increased the Fe levels in the Palouse soil, and all other treatments reduced the extractable Fe. Autoclaving generally decreased extractable Fe in the three soils. Cheng and Ouellette (1970) reported that steam sterilization increased exchangeable Mn in soils low in exchangeable Fe, but did not affect Mn levels in soils with high Fe content. Recently, Sims (1986) has shown that, in soils with pH values >5.2, the dominant Mn forms are organically complexed and Fe-oxide bound, which further supports the interaction of Fe and Mn in the treated soils. Sterilization treatments that resulted in higher pH values would also be expected to decrease Al and Fe levels due to precipitation of hydroxides.

Extractable Nutrient Cations and Microelements

The levels of Ca, Mg, and K in the soils were largely unaffected by the various treatments (data not presented). Cummins (1962) reported that soil irradiation did not affect extractable Ca, Mg, or K. Ferriss (1984) also found that autoclaving did not affect soil concentrations of Ca, Mg, K, and Zn. Skipper and Westermann (1973) showed that extractable Ca levels in three soils they studied were unaffected by propylene oxide, sodium azide, or autoclaving. The same authors also reported that propylene oxide had no significant influence on extractable K. Irradiation or autoclaving did not significantly influence the water-extractable K levels in clay soil studied by Salonijs et al. (1967); however, autoclaving and irradiation increased the Ca and Mg concentrations. As expected, the Na level in each of the three soils was significantly increased by the addition of sodium azide, but none of the other sterilization methods had any effect. Levels of extractable Cu, Zn, Ni, Pb, and Cd were <3.9 mg/kg in the three soils and generally decreased when the soils were autoclaved (data not presented).

Cation Exchange Capacity

The cation exchange capacity of the three soils was not significantly influenced by any of the treatments used. The cation exchange capacities of a sand and a muck studied by Eno and Popenoe (1964) were not significantly altered by steam or gamma radiation sterilization.

CONCLUSIONS

Cobalt-60 irradiation, propylene oxide, mercuric chloride, and autoclaving 2× or 3× were effective means of eliminating the microbial populations of the three soils studied. None of the treatments that eliminated the microbial population significantly influenced the cation exchange capacity values, but measured surface area was significantly reduced by treatment with propylene oxide. Soil pH was increased by treatment with propylene oxide and sodium azide. Extractable Mn was increased when the soils were autoclaved, oven-dried, cobalt-60 irradiated, or treated with sodium azide.

Of the various treatments tested, all of the effective sterilization procedures except mercuric chloride resulted in substantial physical or chemical changes in the soils studied. However, mercuric chloride is highly toxic to humans and care must be exercised when using such a compound. Safe disposal of the treated soil must also be practiced.

Thus, caution should be used in selecting a given sterilization treatment when evaluating degradation and/or adsorption of labile organic compounds in soil. These results indicated that mercuric chloride appears to be the best overall method for pretreatment of soils to be used in adsorption studies involving radiolabeled labile organic chemicals and the batch equilibration technique.

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